

Group A Diary — 05.05.2026 - 12.05.2026

Backend Team: Andrejs Grisin, Vitalina Alipova, Anastasia Petrova, Alan Kurilov

Frontend Team: Igor Sieradzki, Anca Morosan, Arushee Kumar, Kunal Sharma

1. Backend Team

Significant progress was made in the backend architecture and core business logic implementation.

Anastasia extended the database architecture substantially. In addition to the previously existing users and trades models, new database models were introduced for holdings, stock_price_history, net_worth_history, and auto_trades, providing a more complete foundation for portfolio and trading functionality.

The database connection setup was refactored to improve maintainability and security. Instead of hardcoded values, the database URL is now loaded through environment variables using a centralized configuration module (config.py). A dedicated dependencies.py file was also introduced to manage database sessions through the get database dependency, improving modularity and consistency across the application.

A complete authentication system was implemented. The authentication router now provides:

- POST /auth/register for user registration
- POST /auth/login for user authentication

Security features were applied by hashing passwords with bcrypt and generating JWT tokens upon successful login. The authentication flow was structured using Pydantic schemas for request validation and response serialization.

Additionally, environment-based configuration was introduced via a “.env file”. Sensitive credentials such as DATABASE_URL and SECRET_KEY were externalized and excluded from version control. Alembic was initialized to support database schema migrations. Anastasia reviewed Vitalina’s Pull Requests this week.

Vitalina continued backend business logic development by integrating the trading functionality with the database. The buy and sell endpoints now validate available balance, persist trades, and update user balances accordingly. Portfolio tracking logic was expanded with holdings management and bankruptcy checks. Code cleanup and refactoring were also completed. Vitalina reviewed most of the backend code this week.

Further backend endpoints implemented include:

- GET /portfolio — returns all user holdings with:
 - current stock price
 - current position value
 - cost basis
 - profit/loss (absolute and percentage)
 - aggregated totals including portfolio value, total cost basis, total P&L, and cash balance
- GET /trades/history — returns paginated trade history

Andrejs implemented the GET /leaderboard endpoint (Backend Task 4), enabling user ranking functionality. Andrejs also reviewed security implementations.

Alan implemented endpoints for retrieving stock price data and historical stock market information.

2. Frontend Team

Igor set up the frontend project using Vue 3 and established the initial application infrastructure. This included:

- Axios API client configuration with JWT interceptor (src/api/axiosInstance.js)
- Pinia authentication store with login/logout functionality (src/stores/auth.js)
- project setup instructions for local development

The frontend can now be started using:

1. git pull

2. cd frontend
3. npm install
4. npm run dev

Anca implemented the authentication user interface, including:

- /register page
- /login page
- related reusable frontend components

3. Planned Work for the Coming Week

Several frontend features are scheduled for implementation next week (13.05.2025 – 15.05.2025).

Kunal will work on the stock detail page:

- /stocks/:ticker
- stock price chart integration using vue-chartjs
- selectable time periods (1W / 1M / 1Y)
- trading form with live total calculation
- confirmation modal before order execution

Arushee will implement:

- net worth history area chart using vue-chartjs
- /trades page with trade history table and buy/sell filtering
- leaderboard enhancement with highlighting of the currently logged-in user

Summary

This week our group focused primarily on establishing the technical foundation of the application. Major backend features included authentication, database architecture expansion, trading integration, and portfolio tracking. On the frontend side, the project structure and authentication workflows were successfully established.